

Topic 1.1 - Change in Tandem

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Describe how input and output values vary together, including sign and direction on intervals.
- Connect verbal, numerical, graphical, and symbolic representations of a function.
- Construct and critique contextual graphs with justified axes, units, anchors, domain, range, and phases.

Section A: Foundation and representation

Question 1

Foundation

No calculator

Determine whether the relation is a function of x .

x	-3	-1	0	2	2	5
y	4	4	1	-2	3	-2

Question 2

Representation

No calculator

A function has pairs $(-4, 6)$, $(-1, 0)$, $(2, -3)$, $(7, 0)$. State the displayed domain and range, find $f(2)$, and solve $f(x) = 0$ from the display.

Section A continued

Question 3

Reasoning

No calculator

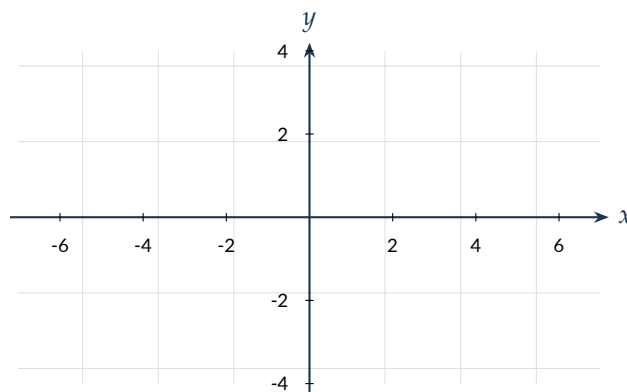
On $1 \leq x \leq 5$, a function is positive and decreasing. Write a complete inequality comparing $f(1)$, $f(3)$, $f(5)$, and 0.

Question 4

Modeling

No calculator

A cyclist is 0 km from school at 7:00, reaches 6 km at 7:20, stays there until 7:30, and returns by 7:42. Draw a labeled graph and identify the segment with greatest rate magnitude.



Question 5

Connections

No calculator

For $f(x) = x^2 - 4$ and $g(x) = |x| - 2$, compare their signs at $x = 1$ and $x = 3$. Why does agreement in sign not make the functions identical?

Section B: Application and reasoning**Question 6**

Algebra

No calculator

If $h(x) = 3x + k$ and $h(-2) = 11$, find k , the preimage of 2, and the direction of h .

Question 7

Piecewise

No calculator

Let $p(x) = 2x + 5$ for $x < 1$ and $p(x) = 10 - x$ for $x \geq 1$. Find $p(-3)$, $p(4)$, and every solution of $p(x) = 7$, respecting piece conditions.

Question 8

Context

No calculator

A cafeteria model $R(n)$ gives revenue from n meals. Identify independent and dependent variables and give a reasonable domain if at most 480 meals can be prepared.

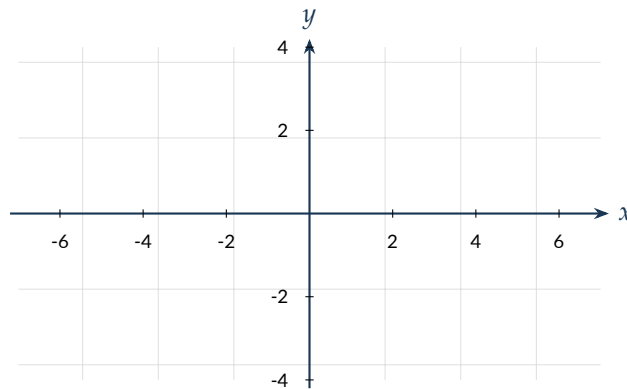
Section C: Challenge and error analysis

Question 9

Challenge

No calculator

Construct two continuous functions on $[0, 4]$ through $(0, 1), (2, 5), (4, 1)$, one with one local maximum and another with at least two.



Question 10

Error analysis

No calculator

A student says, "Since $f(-2) = -7 < 0$, the function is decreasing at $x = -2$." Diagnose the error.

Question 11

Table reasoning

No calculator

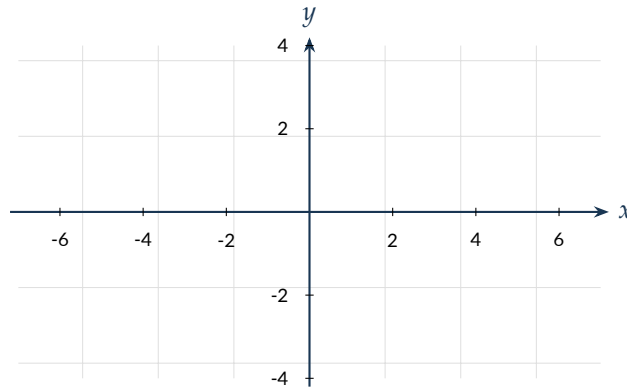
The table gives $m(0) = 5, m(1) = 2, m(2) = -1, m(3) = -1, m(4) = 4$. Describe displayed direction and sign and identify displayed zeros.

Question 12

Challenge

No calculator

Create a function with domain $[-2, 4]$, range $[1, 7]$, decreasing on $[-2, 1]$, constant on $[1, 2]$, and increasing on $[2, 4]$.



Section D: AP-style multiple choice

MCQ 1

No calculator

Which relation defines y as a function of x ?

- A. $\{(-1, 2), (0, 3), (-1, 4)\}$
- B. $x^2 + y^2 = 9$
- C. $y = \sqrt{x + 2}$ for $x \geq -2$
- D. $x = y^2$ for all real y

MCQ 2

No calculator

If $T(d)$ is travel time in minutes for distance d km, what does $T(6) = 42$ mean?

- A. The route is 42 km after 6 min.
- B. A 6 km route takes 42 min.
- C. Time increases 42 min/km.
- D. A 42 km route takes 6 min.

MCQ 3

No calculator

Which description is logically possible on $[0, 5]$?

- A. Negative with all outputs greater than zero.
- B. Positive and decreasing from 12 to 4.
- C. Increasing with $f(0) > f(5)$.
- D. Constant with two outputs at $x = 3$.

MCQ 4

No calculator

$S(t)$ gives students in school t hours after 6 a.m. Which is a reasonable one-day domain?

- A. All real numbers
- B. All integers
- C. $0 \leq t \leq 12$
- D. $S(t) \geq 0$

AP-style free response 1

Calculator permitted - 6 points

A greenhouse records temperature $F(t)$, in degrees Celsius, t hours after 6 a.m.

t	0	2	5	8	11
$F(t)$	22	29	37	34	26

Also $G(x) = 0.4x + 3$.

A. Find $G(F(2))$ and show the intermediate value.

B. Find every displayed preimage of an output greater than 35.

C. Describe displayed covariation and a possible maximum.

D. State a reasonable domain and one limitation of the table.

AP-style free response 2

No calculator - 6 points

A reservoir contains 900 m^3 at $t = 0$. It decreases steadily to 540 at $t = 3$, remains unchanged until $t = 5$, then decreases at a faster constant rate and is empty at $t = 8$.

A. Draw a labeled graph of $V(t)$.

B. Write a piecewise linear rule.

C. State range and direction intervals.

D. Explain why $V(t) > 0$ does not imply increasing behavior.
